

FairLend

Loan Approval Prediction using Random Forest | Team SentiNULLS

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Project Overview

FairLend is a machine learning-based loan approval prediction system designed to assist financial institutions in making accurate, fair, and efficient lending decisions. The model analyzes applicant demographics, income, employment history, credit behavior, and loan-related attributes to determine the likelihood of loan approval. By automating the screening process, FairLend reduces manual effort, minimizes risk, and ensures consistency in credit evaluation. The solution promotes data-driven decision-making while helping lenders improve operational efficiency and customer experience.

Dataset

The project uses a Loan Approval Dataset containing 45,000 applicant records and 14 important features. These features include Person Age, Gender, Education Level, Annual Income, Employment Experience, Home Ownership Status, Loan Amount, Loan Intent, Interest Rate, Loan-to-Income Ratio, Credit History Length, Credit Score, and Previous Loan Defaults. The target variable is `loan_status`, which indicates whether a loan application is approved or rejected.

Data Preprocessing

- Data cleaning and validation of applicant information.
- Handling categorical variables using label encoding techniques.
- Feature scaling applied to important numerical attributes.
- Balancing class distribution to improve prediction performance.
- Splitting the dataset into training and testing sets using an 80:20 ratio.

Model Development

Random Forest Classifier was selected due to its robustness, interpretability, and ability to handle large datasets with complex patterns. The model constructs multiple decision trees and combines their predictions through majority voting. This ensemble-learning approach reduces overfitting, improves generalization, and delivers highly reliable classification results. Hyperparameter tuning was performed to maximize predictive performance.

Model Evaluation

The model was evaluated using several classification metrics including Accuracy Score, Confusion Matrix, Precision, Recall, F1-Score, and Feature Importance Analysis. These metrics helped assess model effectiveness, reliability, and predictive power. The Random Forest model achieved an accuracy of 92.5%, correctly classifying approximately 925 out of every 1,000 loan applications.

Results

The analysis revealed that Credit Score was the strongest predictor of loan approval. Previous loan defaults significantly decreased approval probability, while Annual Income and Employment Experience positively influenced approval chances. Loan-to-Income Ratio was another critical factor, with higher ratios generally leading to increased rejection rates. The model successfully captured these relationships and produced highly accurate predictions.

Business Impact

- Automates the initial loan screening process.
- Improves risk assessment and credit evaluation.
- Reduces Non-Performing Assets (NPAs).
- Promotes fair and unbiased lending decisions.
- Supports regulatory compliance and transparency.
- Accelerates loan approval workflows and customer service.
- Enables data-driven lending strategies.

Conclusion

FairLend demonstrates the successful application of machine learning in the financial sector. By leveraging a Random Forest Classifier, the project achieved high prediction accuracy while maintaining reliability and scalability. The system helps financial institutions make smarter lending decisions, reduce operational risks, and improve overall customer satisfaction. The project highlights the growing importance of artificial intelligence and predictive analytics in modern banking and finance.

Project Summary

Team Name: SentiNULLS

Project Name: FairLend

Model Used: Random Forest Classifier

Dataset: Loan Approval Dataset (45,000 × 14)

Target Variable: loan_status

Accuracy Achieved: 92.5%

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